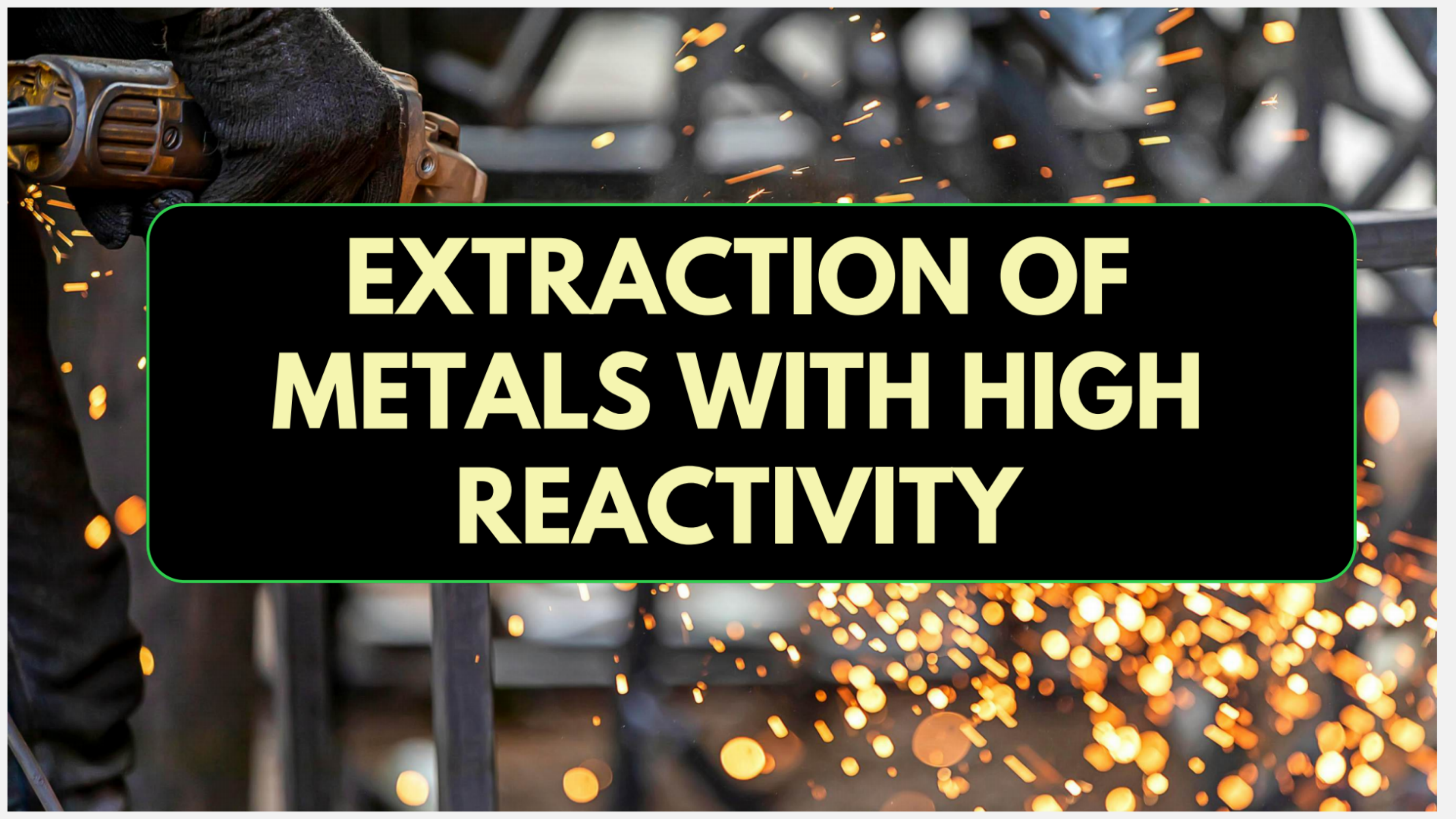


L6

CLASS X - SCIENCE

METALS AND NON-METALS

PRASHANT KIRAD

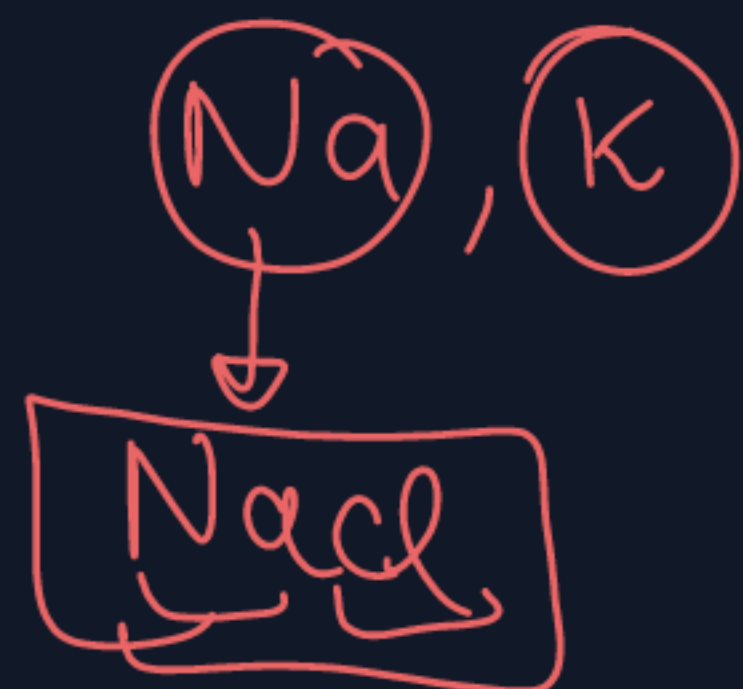


EXTRACTION OF METALS WITH HIGH REACTIVITY

Extraction of High Reactive Metals

The molten compound is electrolyzed to obtain the pure metal.

- Potassium (K)
- Sodium (Na)
- Calcium (Ca)
- Magnesium (Mg)
- Aluminum (Al)

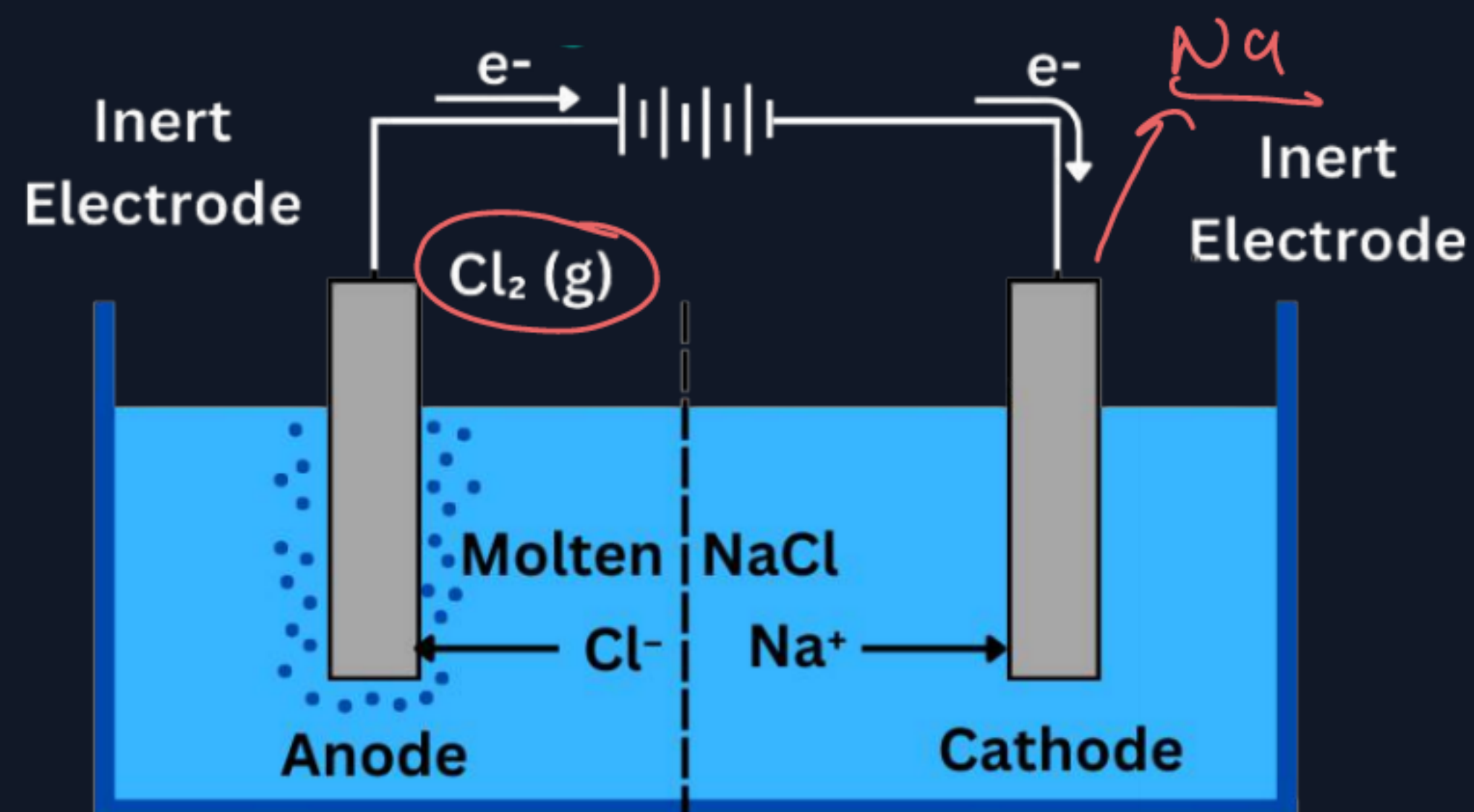
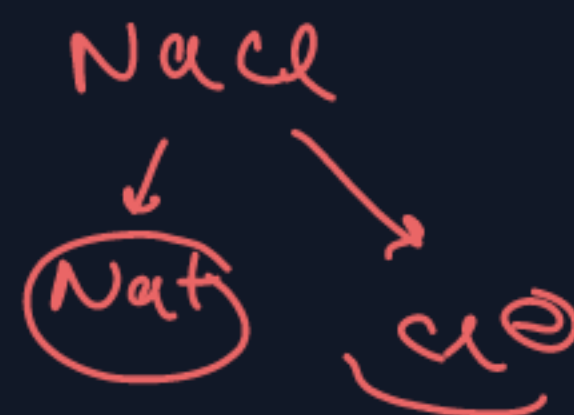


Handwritten note: Molten NaCl (underlined).

Electrolysis of molten NaCl

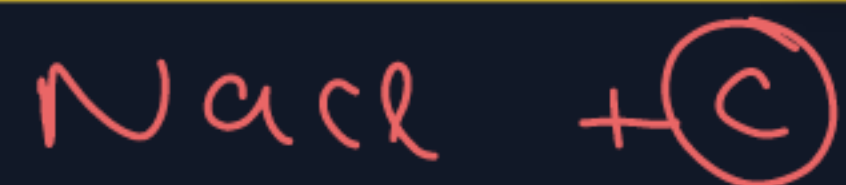
At Cathode (-): $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

At Anode (+): $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$



- ✓ Sodium metal is deposited at the cathode.
- ✓ Chlorine gas is released at the anode

Extraction of High reactive Metals

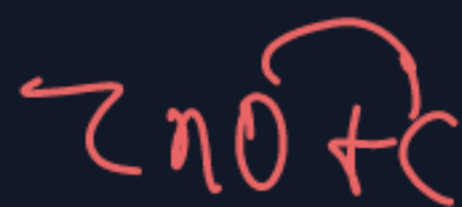


remember

(Why cannot carbon be used?)

These metals are highly reactive and cannot be obtained by heating their oxide with carbon.

They have a greater affinity for oxygen than carbon.



REFINING OF METALS



- Metals obtained by reduction processes are not pure → contain impurities.
- Electrolytic Refining is the most widely used method.

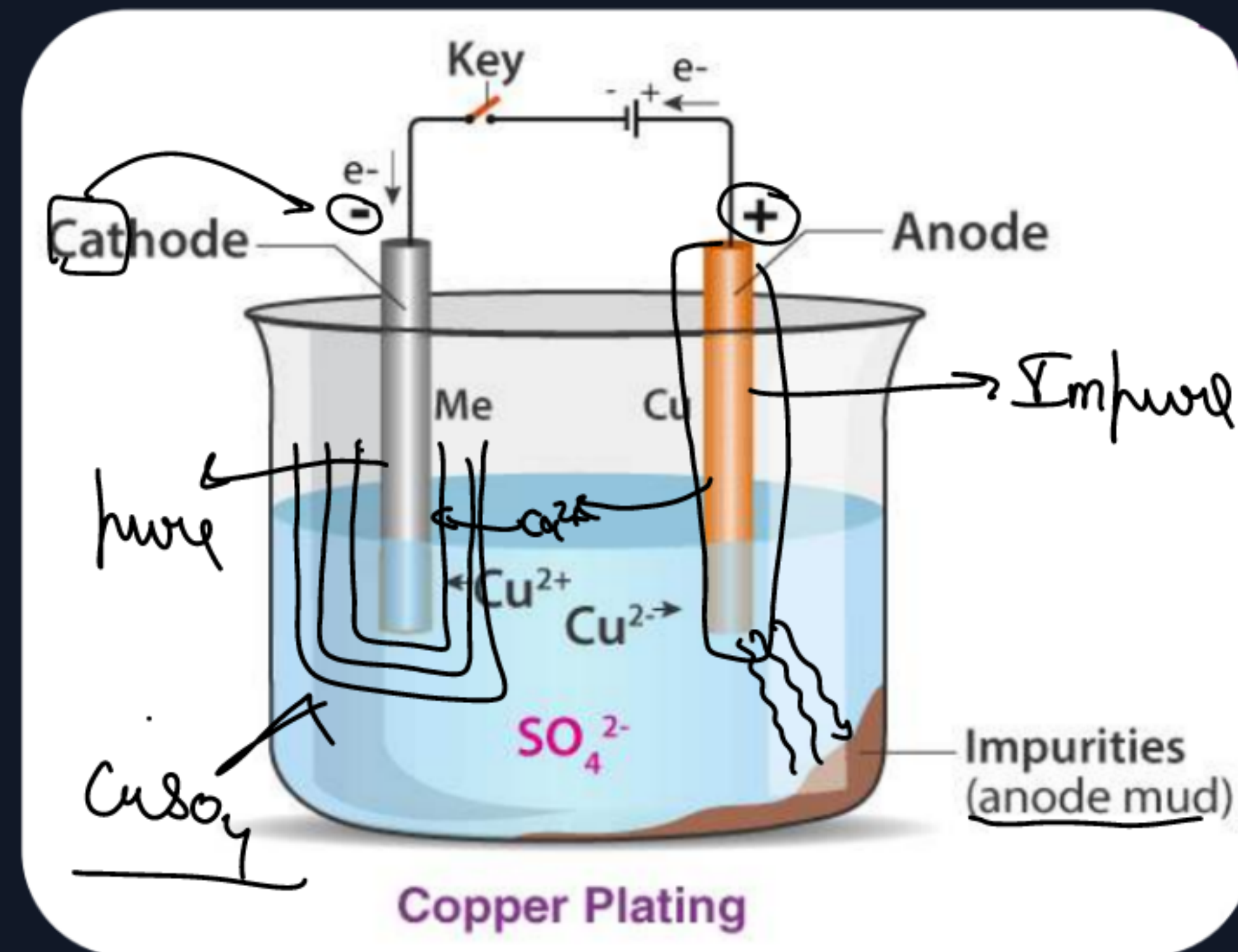
Purification of Metal

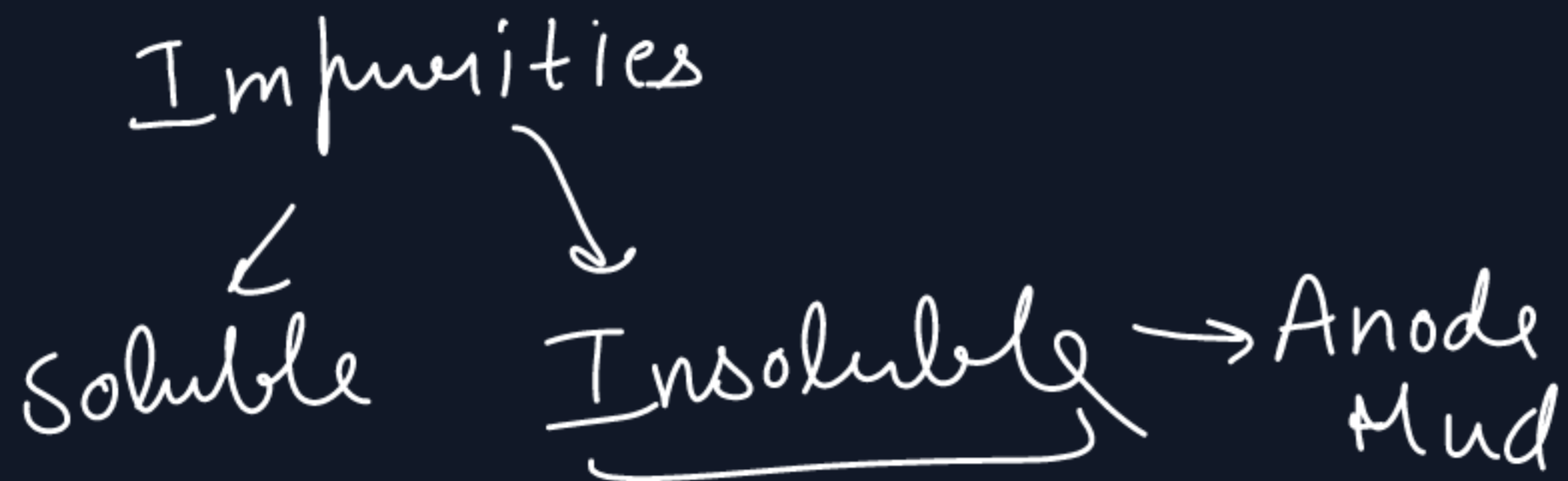
Process (Electrolytic Refining):

At Anode (+): Oxidation: $\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

At Cathode (-): Reduction: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$

- ✓ Anode (+) → Impure metal
- ✓ Cathode (-) → Thin strip of pure metal
- ✓ Electrolyte → Solution of the metal salt (e.g., CuSO_4 for copper).





Working:

On passing current:

- ✓ • Pure metal from anode dissolves into electrolyte.
- ✓ • Equivalent pure metal gets deposited at cathode.
- Soluble impurities → remain in solution.
- Insoluble impurities → settle at bottom of anode as anode mud.

Examples of metals refined electrolytically: Copper, Zinc, Tin, Nickel, Silver, Gold.

Check yourself

(T)

QUES Assertion (A) : Mercury is extracted by heating its oxide, whereas sodium cannot be obtained by heating sodium oxide.

Reason (R): The extraction method depends on the reactivity of metal.

- a) ☒ Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

Check yourself

Ques: During roasting, the evolution of a pungent gas indicates that the ore is most likely:

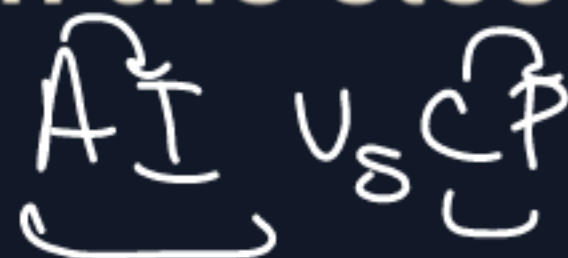
- a) Oxide
- ☒ b) Sulphide
- c) Carbonate
- d) Native



Check yourself

Ques: The metals obtained after reduction are generally impure and are purified by electrolytic refining. In the electrolytic refining of copper:

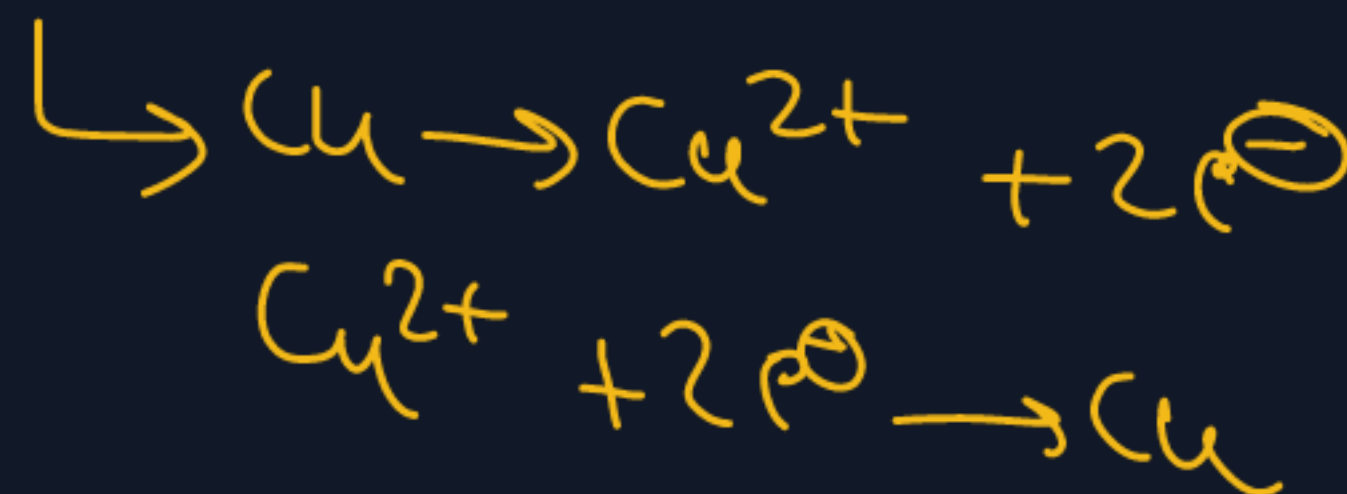
(a) What is used as the anode?



(b) What is used as the cathode?

(c) Name the electrolyte used.

(d) What happens at the anode and cathode when electric current is passed?



(e) What is anode mud?



Check yourself

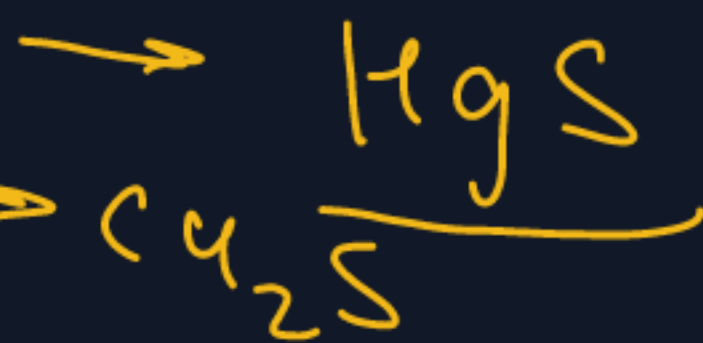
QUES: (2016, 2020, 2025)

State, giving reason, the process used to obtain each of the following metals from their compounds:

(i) Mercury

(ii) Copper

(iii) Sodium



Check yourself

Q. Carbon cannot reduce the oxides of sodium, magnesium and aluminium to their respective metals.

- (a) Why can carbon not be used for their reduction?
- (b) Where are these metals placed in the reactivity series?
- (c) How are these metals obtained from their compounds?
- (d) Explain the extraction of one such metal with suitable electrode reactions.

→ NaCl



Check yourself

- QUES: A student labels an electrolytic refining setup
- ✗ Pure metal connected to positive terminal.
 - ✗ Impure metal connected to negative terminal
 - ✓ Metal salt solution used as electrolyte

Identify the incorrect labels and explain how the setup should actually be arranged

Answer: Impure metal - connected to positive terminal (anode).
Pure metal—connected to negative terminal (cathode)
electrolyte - solution of a soluble salt of the same metal



Check yourself

Q. Two ore samples, labelled as 'P' and 'Q', were selected for processing in order to extract the respective metals from them. Ore 'P' releases CO_2 upon heating, whereas ore 'Q' releases SO_2 . On the basis of the given information answer the questions that follow:

- (A) Identify what kind of ore is 'P'. *→ carbonate*
- (B) Identify what kind of ore is 'Q'. *→ Sulphide*
- (C) Mention the steps involved in the extraction of a metal from ore 'P'.
- (D) Mention the steps involved in the extraction of a metal from ore 'Q'.
- (E) Explain the first step employed in the extraction of ore 'Q'.

(A) Ore P is a carbonate ore (gives CO_2 on heating).

(B) Ore Q is a sulphide ore (gives SO_2 on heating).

(C) Steps for extraction from ore P (carbonate ore):

1. Calcination (heat in limited/absence of air) $\rightarrow$ carbonate converts to oxide + CO_2
2. Reduction of metal oxide (using C/CO) $\rightarrow$ metal obtained
3. Refining (purification)

(D) Steps for extraction from ore Q (sulphide ore):

1. Roasting (heat in presence of excess air) $\rightarrow$ sulphide converts to oxide + SO_2
2. Reduction of metal oxide $\rightarrow$ metal obtained
3. Refining

(E) First step for ore Q (Roasting): Roasting means heating the sulphide ore strongly in excess air to convert it into metal oxide, releasing SO_2 gas.



Check yourself

Hw

Ques: A student draws the following flowchart.

Metal ore $\rightarrow$ Heating $\rightarrow$ Metal oxides $\rightarrow$ Reduction $\rightarrow$ Pure metal.

The student claims every metal can be extracted using this sequence

Evaluate the claim. Support your answer with one example where the statement is correct and one where it is incorrect.

The student's claim is incorrect. Not all metals can be extracted using the sequence.

Metal ore $\rightarrow$ Heating $\rightarrow$ Metal oxide $\rightarrow$ Reduction $\rightarrow$ Pure metal

- This method is suitable only for metals of medium reactivity, whose oxides can be reduced using carbon or carbon monoxide.
- Highly reactive metals cannot be extracted by this method and are obtained by electrolysis.

Example where the statement is correct:

- Iron (Fe): Iron is extracted from hematite (Fe_2O_3) by reducing iron oxide with carbon monoxide in a blast furnace.

Example where the statement is incorrect:

- Aluminum (Al): Aluminum is extracted from aluminum oxide (Al_2O_3) by electrolysis because it cannot be reduced by carbon.

Check yourself

HW

QUES. A student studies the extraction of metals from three different ores/compounds, P, Q and R.

- P: On heating in the presence of excess air, a pungent-smelling gas is evolved.
- Q: On heating in the absence or limited supply of air, carbon dioxide gas is evolved.
- R: The metal associated with R is highly reactive, and its compound cannot be reduced using carbon.

Answer the following:

- Identify the metallurgical processes used for P and Q.
- Name the pungent-smelling gas evolved from P.
- Why is ore P converted into its oxide before reduction?
- Suggest the most suitable method for extracting the metal from R. Give a reason.

Answer: (a) P → Roasting

Q → Calcination

(b) Gas evolved from P → Sulphur dioxide (SO_2)

(c) Sulphide ores are converted into metal oxides because metal oxides are easier to reduce than metal sulphides.

(d) R → Electrolytic reduction (electrolysis)

Reason: Highly reactive metals cannot be reduced using carbon, so they are extracted by electrolysis of their molten compounds.

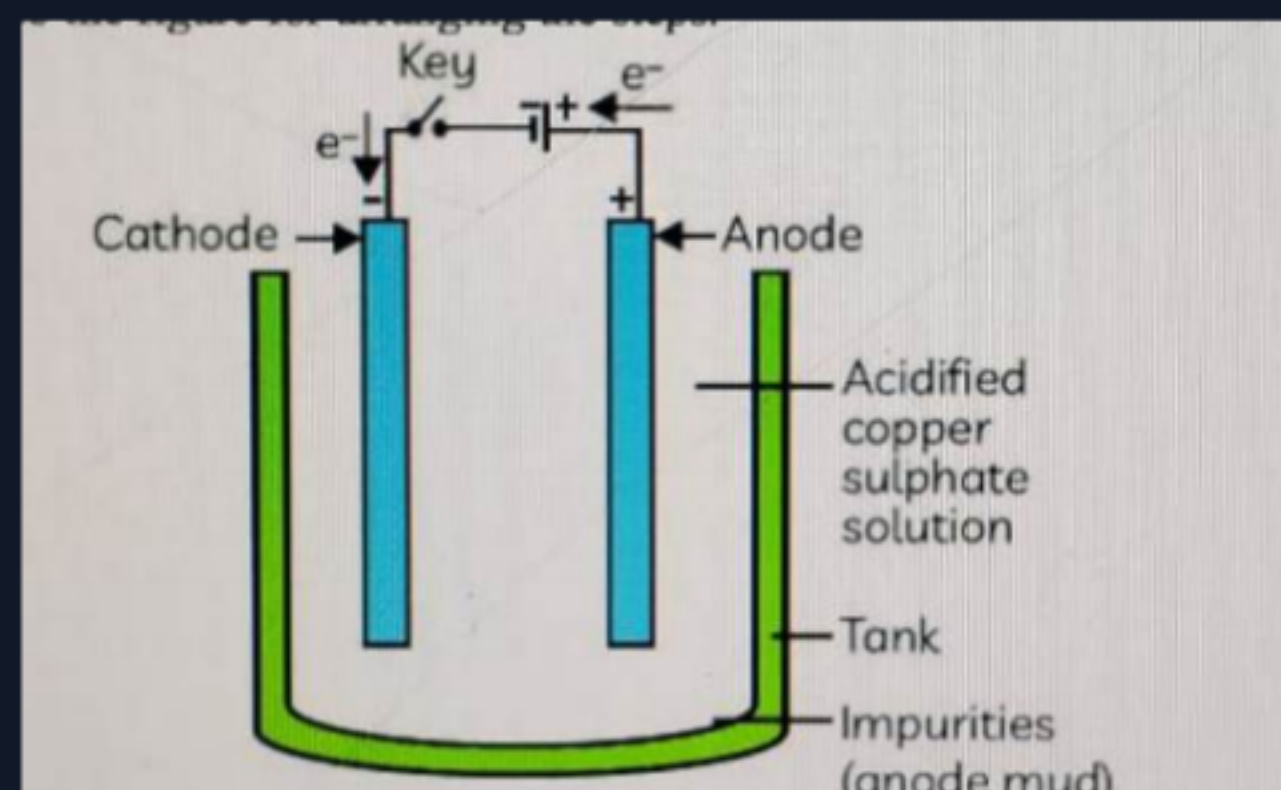


Check yourself

Question: List the process that explains how pure metals are obtained from impure samples by electrolytic refining.

1. Keep impure metal at anode and pure metal at cathode.
2. pass current in the electrolytic solution
3. Insoluble impurities settle in the bottom of the anode mud.
4. Pure metal from the anode dissolves in the solution and deposits on the cathode.

1243



Which option is correct for the arrangement of these steps in the appropriate order?

- a) 1,3,4,2
- b) 1,2,3,4
- c) 1,2,4,3
- d) 1,4,3,2



Check yourself

Ques: A newly discovered metal is being extracted from its ore. Before choosing between carbon reduction and electrolysis, the scientist determines the position of the metal in the reactivity series.

- a) Why is the position of the metal in the reactivity series important?
- b) Which method will be used if the metal is moderately reactive?
- c) Which method will be used if the metal is highly reactive? Give reason.

- (a) The extraction method depends on the reactivity of the metal.
- (b) Moderately reactive metals → Reduction of metal oxide using carbon/CO.
- (c) Highly reactive metals → Electrolytic reduction of molten compounds, because their oxides cannot be reduced by carbon.



Check yourself

QUES: A metallurgist is given three metal oxides for extraction

X < Z < Y

X →

Y →

Z →

Metal oxide	Observation on heating
X_2O	Decomposes directly to metal
Y_2O_3	Does not decompose even at very high temperature
ZO	Can be reduced by carbon

X_2O_3

1. Which oxide belongs to the least reactive metal? Give a reason.
2. Which oxide is most likely extracted by electrolysis?
3. Which oxide can be extracted by carbon reduction?
4. Arrange the corresponding metals in increasing order



Corrosion

Corrosion is the gradual deterioration of a metal due to its reaction with substances such as air and moisture.

A common example is the rusting of iron, which occurs when iron is exposed to air and moisture, forming a reddish-brown coating of hydrated iron(III) oxide, **$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$** .

Due to corrosion, the metal's lustrous surface fades away.

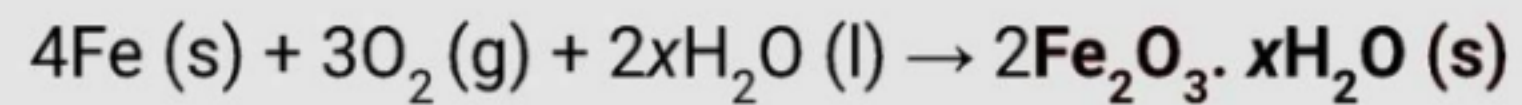
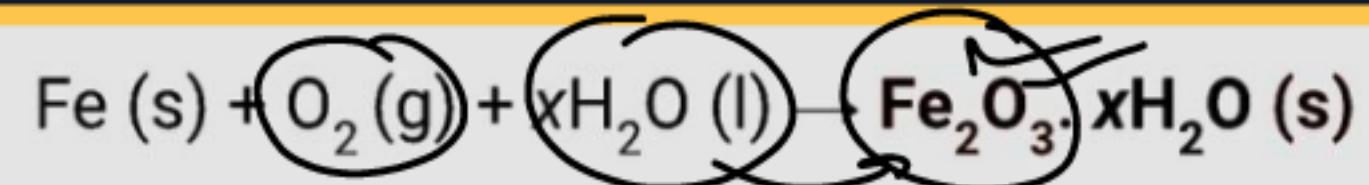


Metal shine fades away after corrosion

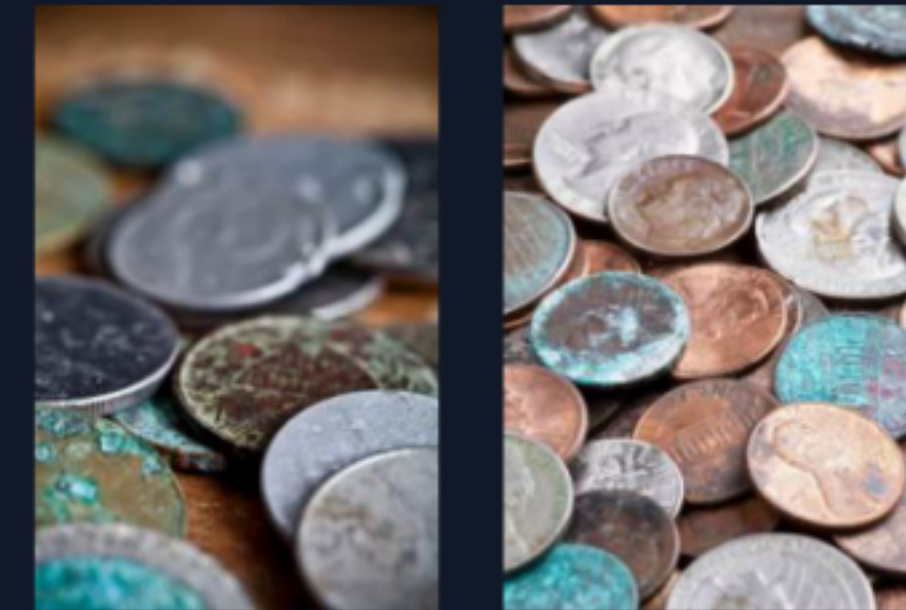
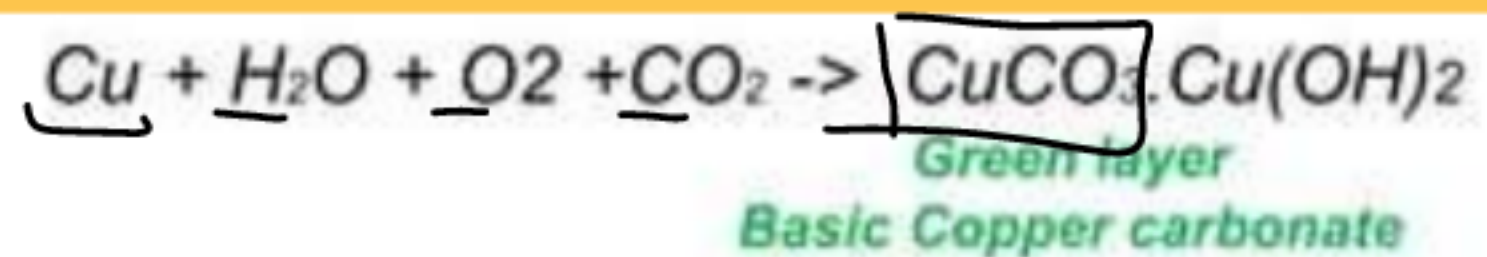


Metal shine Before corrosion

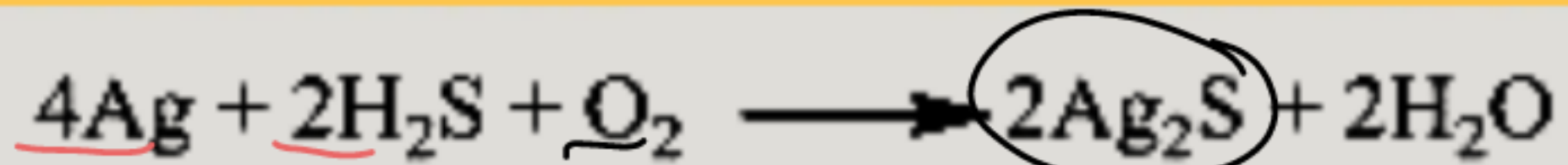
RUSTING OF IRON : When iron comes in contact with moist air, it forms hydrated ferric oxide. This oxide has a reddish-brown color.



Tarnishing of Copper : Copper reacts with moist carbon dioxide from the air to produce copper carbonate. This reaction results in a green layer forming on its surface.



Tarnishing of Silver : Silver reacts with hydrogen sulphide in the atmosphere to form silver sulphide, which gives it a black coating.





Advantages of Corrosion

Patina (Advantageous Corrosion):

- Corrosion can sometimes be beneficial. For example, copper develops a protective green layer called patina after oxidation.
- This layer acts as a barrier that prevents further corrosion.

Patina ←



PREVENTION OF CORROSION

Prevention of Corrosion

To stop metals like iron from rusting, three primary techniques are used

*Using a
protective
barrier*

*Sacrificial
protection*

Alloying

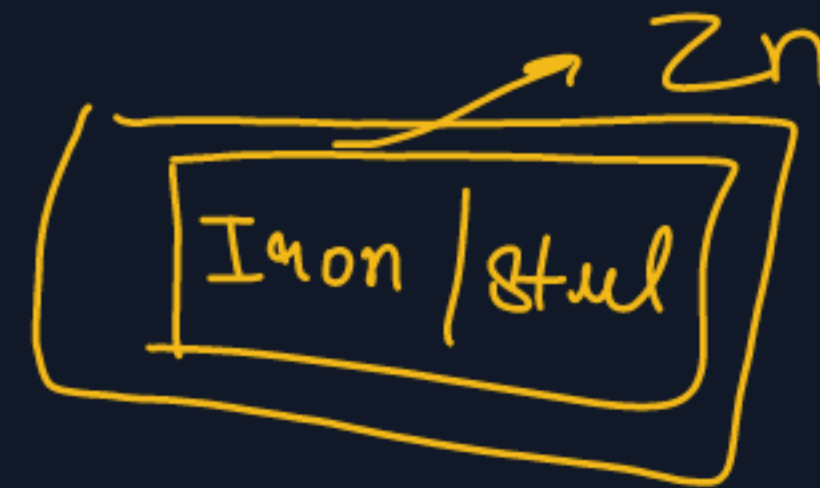


Barrier Protection:

This approach involves placing a protective coating over iron to block rust-causing agents like air and water vapor.

Common methods include:

- i. Painting the surface
- ii. Lubricating with oil or grease



Sacrificial Protection:

This technique uses a more reactive metal to shield a less reactive one.

The reactive metal corrodes instead of iron, thus offering protection.

Galvanization: A technique where a thin zinc layer is applied to steel or iron. Even if the coating breaks, the zinc still protects against rust.

Sheets of galvanized iron are commonly used to make: Water buckets, S drums and Roofing sheets



Alloying

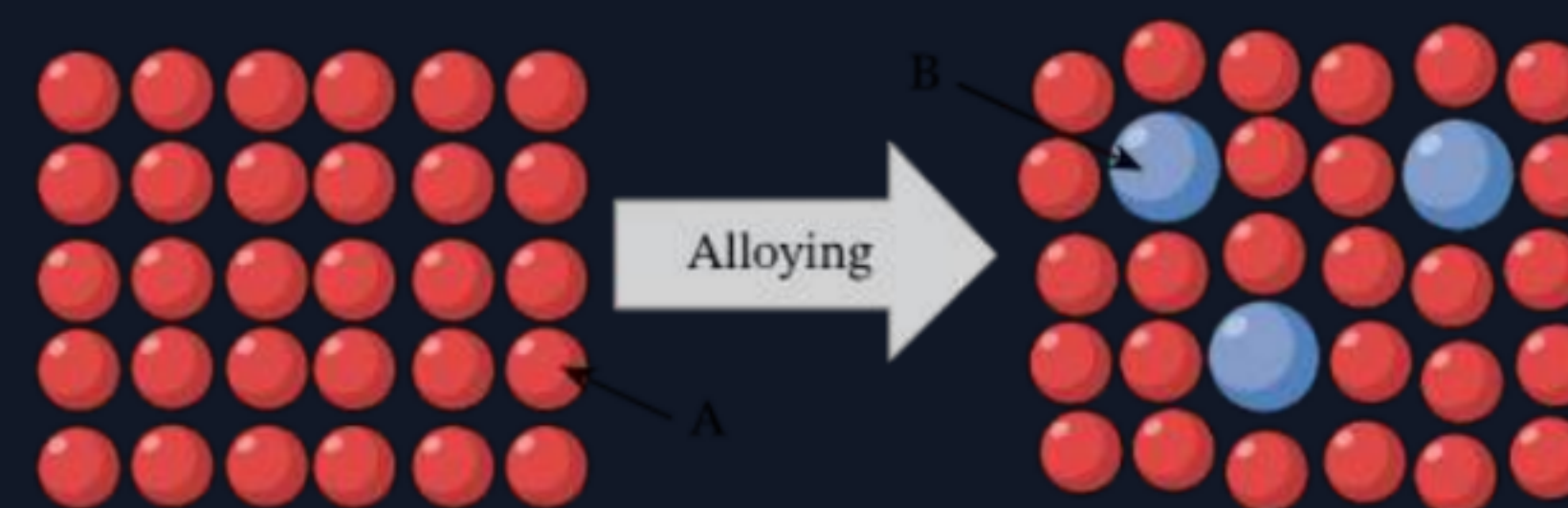
✓ Alloy is a homogeneous mixture of two or more metals, or a metal and a non-metal.

- **Preparation:** The primary metal is first melted, then other elements are dissolved in definite proportions and cooled to room temperature.



Properties of Alloys:

- ✓ Electrical conductivity less than pure metals.
- ✓ Melting point lower than pure metals.
- ✓ Mechanical strength and corrosion resistance often greater than pure metals.



- ✓ **Amalgam**: If one of the metals in an alloy is mercury,
- ✓ **Brass (Cu + Zn)**: Harder than copper, corrosion resistant, and lower electrical conductivity than copper. "CoZ"
- **Bronze (Cu + Sn)**: Hard, strong, corrosion resistant, and lower electrical conductivity than copper. "CUT"
- ✗ **Steel (Fe + C)**: Harder and stronger than iron. "FeCu"
- ✓ **Stainless Steel (Fe + Cr + Ni)**: Hard, strong, shiny, and highly resistant to corrosion/rusting. "FirCaNahi"
- ✗ **Solder (Pb + Sn)**: Has a low melting point; used for joining electrical wires/components. "LT"

QUESTIONS



2018, 2020

Q. Why is 22-carat gold used for ornaments instead of 24-carat gold?"

- 24-carat gold is pure gold, but it is very soft and cannot be used for making durable ornaments.
- 22-carat gold is an alloy of gold with small amounts of copper or silver.
- Alloying makes gold harder and more durable, while still keeping it suitable for making jewellery.



QUESTIONS

HW

QUES: Food cans are coated with tin and not with zinc because

- a) Zinc is costlier than tin.
- b) Zinc has a higher melting point than tin.
- c) Zinc is more reactive than tin
- D) zinc is less reactive than tin [NCERT]



QUESTIONS

QUES. Two companies manufacture iron water pipes.

- company A coats the pipes with zinc.
- Company B coats the pipes with copper.

After a few months, both coatings develop scratches. Which company's pipes will show better corrosion resistance? Justify your answer

Answer: Company A, zinc is more reactive than iron, so it oxides first and protects iron from corrosion (sacrificial protection)..

copper is less reactive than iron. If its coating is scratched, the exposed iron corrodes more rapidly.



QUESTIONS

KW

Q. A piece of iron rusts when it comes in contact with air and moisture. Prakash had two identical shiny iron pieces P and Q. To prevent the pieces from rusting, he coated piece P with oil paint and galvanized piece Q with a coat of zinc metal. He noticed that the coatings were not complete and that a small part of the iron was exposed in both the pieces.

What is Prakash likely to observe about the exposed parts of the two iron pieces after some days? Explain why.

Observation after some days:

- Piece P (oil paint, incomplete): The exposed iron part will rust.
- Piece Q (galvanised with Zn, incomplete): The exposed iron part will NOT rust (or rusts much less).

Reason:

- In P, iron is directly exposed to air + moisture → rusting occurs.
- In Q, zinc is more reactive than iron, so zinc sacrifices itself and protects iron (sacrificial protection).

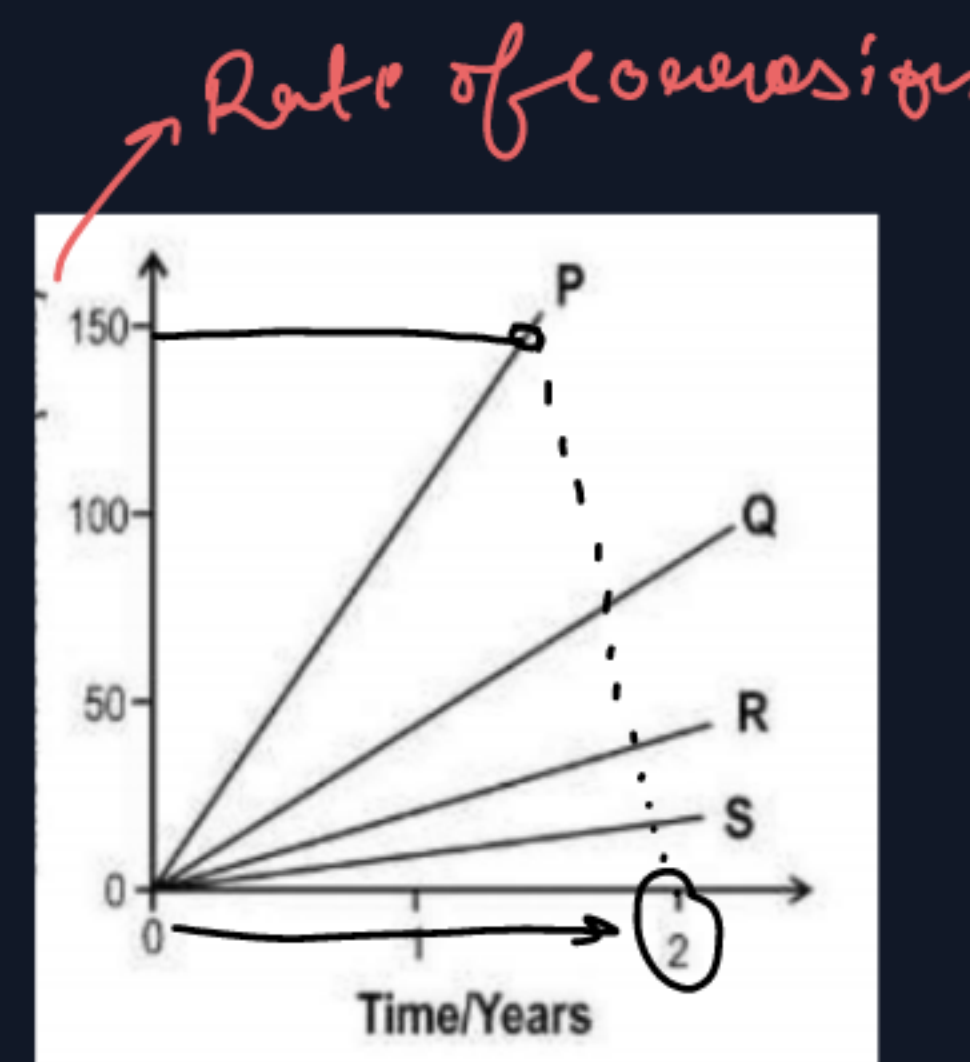
QUESTIONS

Q. The presence of acidic gases in the air increases the rate of corrosion. Furthermore, an increase in temperature can also increase the rate of corrosion. The graph below is created under 4 different conditions (shown below in the table) of temperature and acidic nature of air.

Condition	Temperature (in °C)	pH value
1	20	6
2	30	6
3	30	7
4	20	7

Which of the graphs represents condition 2?

- (a) P
- (b) Q
- (c) R
- (d) S



QUESTIONS

hw

Ques: Read the following :

Iron structures in coastal areas are more prone to rusting because of the presence of moist air. To protect iron from corrosion, it is often coated with zinc.

- (a) Why does iron rust faster in coastal areas?
- (b) Name the process of coating iron with zinc to prevent rusting.
- (c) Why does the zinc coating protect iron even if the coating gets scratched?

Answer: (a) Moist air provides oxygen and water, which are necessary for rusting.

(b) Galvanisation

(c) Zinc is more reactive than iron, so it corrodes in preference to iron and protects the exposed iron from rusting.

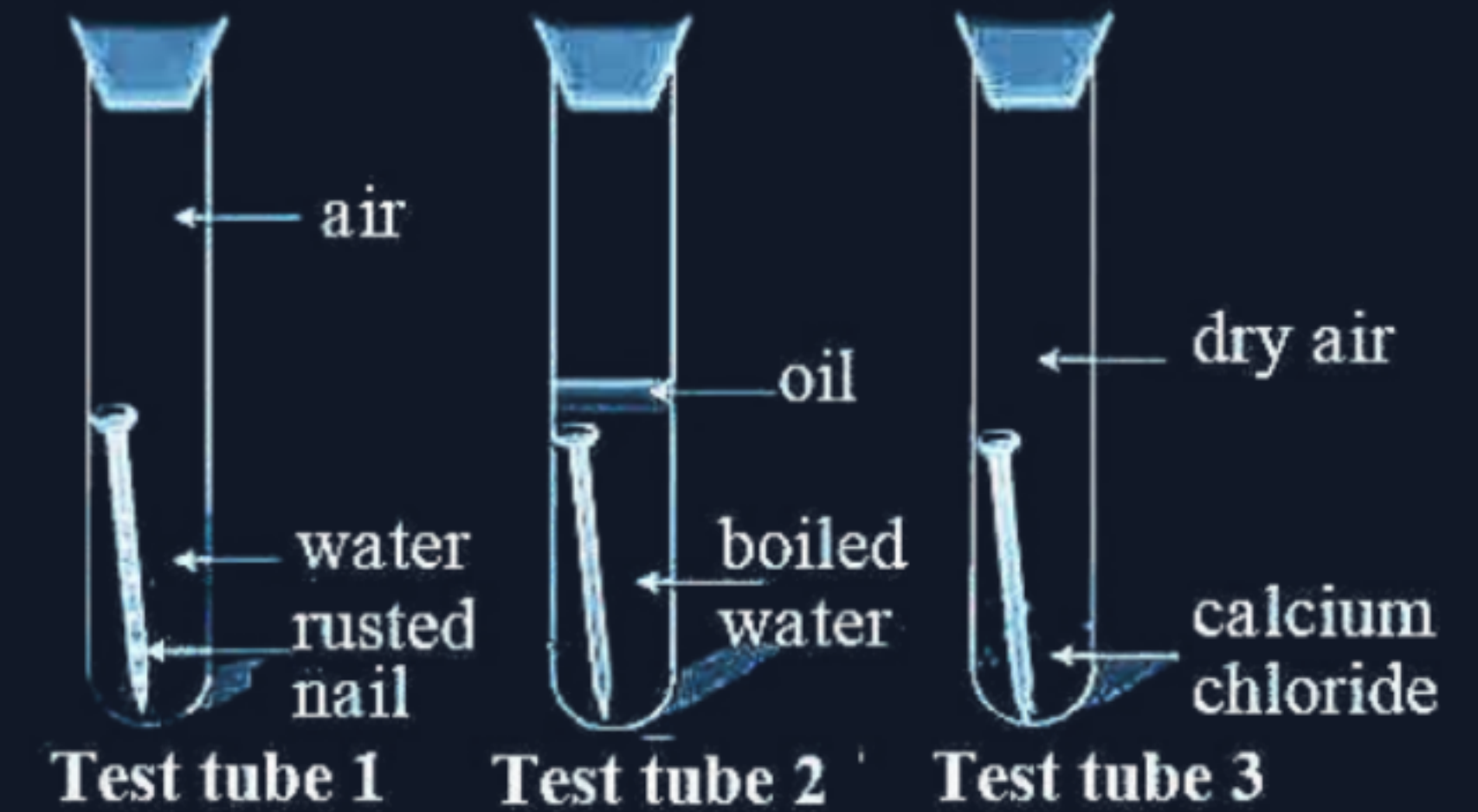


QUESTIONS

HW

Ques: Three experiments to study the process of rusting are given below. Observe the three test tubes and answer the following questions.

- Why is the nail in test tube 2 not rusted?
- Why is the nail in the test tube 1 rusted highly?
- Would the nail in test tube 3 get rusted?



QUESTIONS

HW

QUES: Give reasons for the following:

- i) Silver and copper lose their shine when they are exposed to air. Name the substance formed on the surface in each case.
- ii) Tarnished copper vessels are cleaned with tamarind juice.
- iii) Aluminium is more reactive than iron yet there is less Corrosion of aluminium compared to iron when both are exposed to air.

Ans: i) These metals are corroded. Silver forms black Ag_2S (silver sulphide) and copper form greenish layer of basic copper carbonate..

ii) Tamarind contains acid which reacts with basic copper carbonate and products gets dissolved and removed from copper vessels.

iii) Aluminium forms oxide layer in its surface which doesn't further react with air.

QUESTIONS

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